Fair Drop

Making speed worthless in high-demand ticket sales

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> **The claim in one sentence.** Ticket bots are not breaking the rules of the sale — they are playing them correctly, because the sale rewards speed. We change the clearing rule so that arrival time carries no information, which removes the payoff instead of trying to detect the player.

SECTION 1

The problem, and what we change

Audience: everyone. No prior knowledge assumed.

A concert has 180,000 tickets. 13 million people try to buy them. That is the real situation, and no software fixes it — most people were always going home without a ticket. [1]

So the only question the system actually answers is: **who gets them?**

Today the answer is: whoever's request reaches the server first. That sounds like a queue at a shop, and it sounds fair. It is not the same thing. You are not standing in line with your body. You are sending a message over a network.

- A program can send that message thousands of times per second, from a machine sitting close to the server, with no hand and no eye involved.
- A person taps a phone on ordinary wifi and sends it once, in a few hundred milliseconds.
- So “first come, first served” really means **whoever has the better software and the shorter network path**.

That is not a tie-breaker. It is the entire contest. Fans did not lose to other fans. They lost to scripts — and then bought the same tickets back from those scripts. In the Coldplay Mumbai sale, inventory was gone in about thirty minutes and a ■12,500 ticket was listed on resale above ■3.36 lakh. [1]

What everyone is trying instead

Catch the bots. CAPTCHAs, IP blocking, rate limits, purchase caps, and increasingly the courts: the FTC and seven states sued Live Nation and Ticketmaster in September 2025, South Korea widened its anti-scalping law in January 2026, Chinese regulators summoned twelve platforms in February, and Washington DC passed the RESALE Act in August. [2][3][4]

It is not working. Automated traffic passed half of all web traffic, scalping bots are a large share of it, and LLMs have made writing a bot nearly free. [5] Security vendors now say plainly that CAPTCHA, IP blocking and rate limiting are no longer sufficient on their own. [6]

> **Why detection cannot win.** The bots are not cheating. The sale is a race, and they are fast. Detection is therefore an attempt to punish people for being good at the exact thing the sale was designed to reward. There is no version of that which converges.

What we change

Stop selling by arrival order. Sell in **turns that close together**.

The sale runs as a short series of turns. In each turn everyone submits once, privately, and **it makes no difference when inside the turn you submit**. At the end of the turn every submission is looked at together and the tickets move. Then the next turn opens, with the result of the last one visible to everybody at the same instant.

Now ask what a bot can still do for you:

- Submit faster — no gain. The turn is still open.
- Submit a thousand times — no gain. One entry counts.
- Sit in a data centre next to the server — no gain. Nobody is racing.

> **We did not catch the bots. We made them pointless.** Speed stops being a strategy because there is nothing left to be fast at.

What this does and does not fix

It fixes: who wins. Allocation stops tracking network infrastructure and starts tracking how much someone actually wants the ticket.

It does not fix: scarcity. 13 million people still cannot attend. Nor does it by itself end resale — that needs tickets bound to the buyer, which is a separate and complementary change. We say this plainly rather than overclaiming.

The demonstration

Everyone in the room joins from their phone and gets an agent. **Round one** runs the sale the way it works today: first come, first served. The same few fast agents take almost everything, and the board shows who. **Round two** runs the same participants under turn-based clearing. Different people win.

Same room, same agents, one rule changed.

SECTION 2

The problem as an allocation rule

Audience: engineers

Strip the domain away and a ticket sale is an allocation function: given a set of requests for a scarce good, decide which requests are satisfied.

The current rule

First-come-first-served allocates in ascending order of arrival timestamp at the server, until inventory is exhausted. The selection key is **t_arrival**, and nothing else.

t_arrival is a function of request automation rate, network round-trip time, geographic proximity to the origin, retry aggression, and connection concurrency. **None of these correlate with demand for the good.** The allocator is therefore selecting on infrastructure quality while presenting itself as selecting on eagerness.

> Restated: the current design does not have a bot problem. It has an **objective function problem**. Automation is the optimal strategy against the stated rule, and the observed behaviour is rational play, not circumvention.

Why the detection layer is structurally disadvantaged

Bot detection is a binary classifier over requests, under adversarial drift. Four properties make it a poor fit for this specific workload:

1. **The base rate inverts exactly when you need the signal.** Detection finds anomalies against normal traffic. At an on-sale, all traffic is anomalous — millions of clients arriving in the same second, refreshing identically. The discriminative signal collapses at the only moment it matters.
2. **The adversary has removed the fingerprint.** Residential proxy farms, real devices, aged accounts created weeks earlier, and human click farms. Vendors describe the scalper stack as operating across bulk account creation, queue automation, drop-time request automation and the CDN edge, each requiring a separate defence. [6]
3. **Error costs are asymmetric and the asymmetry is visible.** A false positive is a blocked fan and a public complaint. A false negative is silent. Thresholds are therefore structurally pushed toward permissiveness.
4. **The payoff is large enough to survive a good filter.** With resale multiples in the tens, a 99% effective filter still leaves a profitable operation. Detection must be near-perfect against an adversary that iterates cheaply.

Scale evidence: the FTC complaint alleges an internal review found five brokers controlling 6,345 accounts holding 246,407 tickets across 2,594 events. [2] That is the account inventory sitting in place *before* the sale opens, and it defeats per-account purchase caps by construction.

The problem we are actually solving

Design an allocation rule such that:

- **C1.** The weight of t_arrival on the allocation outcome is zero.
- **C2.** The number of entries a single actor can hold is bounded.
- **C3.** Every participant can observe that the same rule was applied to the same state — the outcome must be checkable, not asserted.
- **C4.** The rule must not reintroduce a speed advantage through a side channel, including the channel by which participants learn the state.

C4 is the one that is easy to miss and is where most naive designs fail. It is addressed in §3.

SECTION 3

Our approach, and whether it is buildable

Audience: engineers

Mechanism: discrete-turn sealed clearing

The sale is divided into N turns of fixed wall-clock duration W. Within a turn, each participant may submit at most one bid. Bids are not visible to other participants during the turn. At the turn boundary a single clearing function runs over all bids received in the window, allocates against remaining inventory, and publishes the turn result.

A participant's policy is three numbers: **floor**, **ceiling**, and **aggression** — how fast to walk from one to the other as turns elapse. A human sets them on a phone; an agent may generate or adapt them. Either way the fast path is arithmetic, and no model inference sits inside a turn.

Why this satisfies the constraints

- **C1 within a turn.** The clearing function is invariant to the order in which bids arrived inside the window. Submitting at 5ms and at W-5ms are indistinguishable outcomes. Latency advantage is zero by construction, not by policy.
- **C2.** One bid per identity per turn is a uniqueness constraint on (sale, turn, identity). Bounding identities themselves is an orthogonal problem — see §4.
- **C3.** Clearing is one serialized write. Every subscriber receives the same resulting state.
- **C4.** This is the interesting one. Between turns, participants adapt on published state: last clearing price, remaining inventory, who is still bidding. If that state reaches participants *unevenly*, whoever learns first gets more wall-clock time to compute the next bid — and the latency advantage returns through the information channel rather than the submission channel.

> **The load-bearing requirement is therefore not fast writes. It is a single serialized clearing write with simultaneous fan-out to every participant.** That is the property SpacetimeDB provides natively: reducers execute serially, and subscribers receive the resulting rows without an application-level broadcast loop.

Honest scoping of the platform claim

The mechanism is not platform-dependent. Turn-based sealed clearing can be implemented on PostgreSQL with a scheduler and a socket tier. We are not claiming impossibility, and we will not claim it to a judge.

The claim we do make: satisfying C3 and C4 conventionally requires a database, a scheduler, a push/fan-out tier, and hand-written guarantees about ordering and simultaneity across them. Here it is tables, reducers and subscriptions, and the guarantee is a property of the engine rather than of our code. **Three services we did not have to build because the database is the server.**

Feasibility

Component	Work	Risk
Clearing function	Sort bids in the turn, prefix-scan against remaining inventory, write allocations. O(n log n) over a small n.	Low
Turn boundary	Scheduled reducer at t+W. Guarded so only the module identity may invoke it.	Low

Component	Work	Risk
Bid submission	One reducer, one uniqueness constraint. Branches on sale mode.	Low
Baseline (queue) mode	Same entrypoint, allocates immediately while inventory remains. This is the round-one comparison and is not optional scope.	Low
Agent processes	Node clients holding identities. Policy is three numbers plus a step function.	Low
Projector view	Subscription over allocations and turn results, grouped by participant class.	Medium — legibility work, not systems work
Phone client	Join, set floor/ceiling/aggression, watch your own agent.	Medium — onboarding under time pressure

Nothing in the critical path makes an outbound network call, so the demo does not depend on a third-party API responding. The whole system runs against a local standalone server over venue LAN if the network disappoints.

Known risks

- **Legibility.** An auction resolving in three seconds is unreadable. Turns are also the fix for this: 8–10 turns at a few hundred milliseconds each is the same total duration but it visibly ticks — price moving, agents dropping out, inventory draining.
- **Optics.** “Auction” can be heard as dynamic pricing, which is a separate and unpopular thing. The pitch leads with *who wins*, never with what the seller earns. The ranking rule inside a turn is a parameter — see §4.
- **Scope.** Two clearing modes must both work, because the contrast is the demonstration. Wallets, tiers, seat maps and real identity are out.

SECTION 4

Alternatives considered

Audience: engineers

Approach	What it does	Why not on its own
Bot detection / WAF Cloudflare, HUMAN, Prosopo	Classifies and blocks automated traffic at signup, queue, drop API and edge.	Fails the four structural properties in §2. Vendors themselves state single-layer defences are insufficient. [6] **Complementary, not competing** — it reduces junk volume; we remove the payoff. A real deployment runs both.
Per-account purchase caps	Limits tickets per identity.	Defeated by bulk aged accounts. 6,345 accounts across five brokers in one complaint. [2] Caps bind identities, and identities are cheap.
Waiting room with randomised queue position	Randomises order among those present at open.	Better than raw FCFS and widely deployed. But you must still be present at the instant of open, and bulk accounts multiply draws. Randomises the race without removing it.
KYC lottery with deposit	Verified one-entry-per-person, funds held pre-draw, random allocation at face value.	Mechanically sound — kills both speed and Sybil entries, and preserves the artist's face price. Rejected here on **platform fit**: a draw is a batch allocation with no contention and no ordering to arbitrate, so it would run identically on a single-node RDBMS with a cron job. Also leaves resale margin intact unless tickets are non-transferable.
Continuous open auction	Live order book, clears continuously.	Removes resale margin but **reintroduces the original failure**: the fastest agent reacts to price movement first. This is precisely the dynamic we are eliminating.
Identity-bound non-transferable tickets	Ticket admits only the buyer.	Solves *resale*, not *allocation*. Orthogonal and composable. Needs a sanctioned at-face transfer path or it breaks legitimate resale.

The ranking rule is a parameter, not a commitment

Inside a turn, once arrival order is discarded, something must rank the bids. This is one comparator and it is worth being explicit that it is a policy choice with different consequences:

- **Price-ranked.** Highest bids win. Eliminates resale margin, because the gap between face price and market price is what scalpers monetise. Consequence: buyers pay closer to market, and the surplus goes to the seller rather than to the scalper — which is defensible economically and unpopular publicly.
- **Random-ranked among qualifying bids.** Preserves the face price and the artist's intended gift to fans. Consequence: scarcity persists, so resale margin persists, and this branch needs non-transferability to be complete.

Both are the same reducer with a different sort. We can demonstrate both and let the room argue about which is fair — a better conversation to have with a panel than defending one of them.

SECTION 5

High-level design

Audience: engineers

Processes

- **Module (TypeScript on SpacetimeDB).** The exchange. Sole authority. Tables and reducers, no external application server.
- **Agent processes (Node).** One per participant, each holding its own identity. Two classes: a *scalper* that polls hard and buys on sight, and a *fan* that behaves like a person. The asymmetry is real, not simulated — the scalper is faster because it genuinely polls harder.
- **React web client.** Two views off the same subscription: phone (your agent, your result) and projector (the room).

Tables

```
sale (id, mode: queue|turn, state, turn_index, turn_ends_at, inventory_remaining)
participant (identity, display_name, class: scalper|fan, floor, ceiling, aggression)
bid (id, sale_id, turn_index, participant, price, qty, seq, state)
allocation (sale_id, turn_index, participant, price_paid, round)
turn_result (sale_id, turn_index, clearing_price, allocated, remaining)
```

Index every column used in a filter or join. **round** and **turn_index** are columns, not schema variants — re-running the demo produces new rows and never a migration.

Reducers

Reducer	Behaviour
open_sale(mode, turns, W)	Sets mode and state, schedules the first turn boundary.
submit_bid(price, qty)	**Single entrypoint, branches on sale.mode.** In queue mode: allocate immediately while inventory remains. In turn mode: record the bid against the current turn and return. This one branch is the entire difference between the two rounds.
close_turn(sale_id)	Scheduled at turn_ends_at. Ranks the turn's bids, allocates against remaining inventory, writes allocations and turn_result, opens the next turn or settles. Guarded on ctx.sender == module identity.
settle(sale_id)	Freezes the round and writes the summary the projector reads.

Turn lifecycle

```
open → bids accumulate (invisible to participants) → scheduled close_turn fires
→ rank, allocate, publish turn_result → fan-out to all subscribers simultaneously
→ agents adapt policy on published state → next turn opens
```

The single serialized close_turn write and its simultaneous fan-out are what satisfy C3 and C4 from §2. Everything else is ordinary CRUD.

Demo topology and fallback

- Room joins by QR. Projector on the big screen, phones in hands.
- Runs against a local standalone server for the entire build. Cut to hosted only for the audience-joins moment, warmed beforehand.
- If the venue network fails: same build, laptop on the venue LAN or a hotspot. No component in the critical path requires the public internet.

What gets cut first

In order: wallets and budgets, price tiers, multiple lots, anything resembling authentication beyond an anonymous identity with a display name. The demonstration needs one lot, two rounds, and a scoreboard.

Build order. Queue mode plus the projector first — that is a complete and demoable half of the story on its own. Turn-based clearing is the second half and is a scheduled reducer plus a sort. A working round one with a described round two still demos; the reverse does not.

REFERENCES

Sources

[1] Coldplay Mumbai on-sale: approximately 13 million users for 180,000 tickets across three shows, sold out in roughly 30 minutes; resale listings including a ■12,500 ticket above ■3.36 lakh. Malay Mail, September 2024.

[2] FTC and seven states v. Live Nation / Ticketmaster, filed 17 September 2025, under Section 5 of the FTC Act and the BOTS Act; complaint cites an internal review finding five brokers controlling 6,345 accounts holding 246,407 tickets across 2,594 events. Covington, Inside Privacy, October 2025. Defendants moved to dismiss on 6 January 2026, characterising the action as agency overreach (Variety, January 2026); the motion remained undecided as of early May 2026 (TicketNews, May 2026). Separately, a jury found unlawful monopolisation of US ticketing and amphitheatre markets on 15 April 2026 (Music Business Worldwide, May 2026).

[3] South Korea expanded its anti-scalping law on 29 January 2026; Beijing regulators met twelve platforms including JD.com, Didi and Tencent on 12 February over train ticket sales, with seven further platforms summoned on 10 April. CNBC, June 2026.

[4] Washington DC RESALE Act, banning bot scalpers and speculative ticket listings. Reynolds Center, August 2026.

[5] Imperva Bad Bot Report 2025: automated traffic reached 51% of total web traffic in 2024, with scalping bots a substantial share; LLM adoption has lowered the barrier to building ticket bots. Summarised by STCLab, April 2026.

[6] Layered anti-scalping defence across signup filtering, queue behaviour, drop API and CDN edge; CAPTCHA, IP blocking and rate limiting described as no longer sufficient alone. Prosopo, June 2026.

Verification note: reference [2] describes litigation that was active and undecided at the time of writing. Confirm the current status of the motion to dismiss before presenting — a ruling would be a materially fresher hook than the filing.